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## 1. The Natural Numbers

### Exercise 1.1

Prove using mathematical induction that for all positive integers  $n$ ,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

*Proof.* Let  $X$  be the set of all natural numbers for which the formula above holds true.  $1 \in X$  because

$$1 = \frac{1(1+1)}{2}.$$

Now assume that  $n > 1$ , and that  $k \in X$  for all  $k < n$ . We know from our induction hypothesis that the formula holds for  $n - 1$ , so

$$1 + 2 + 3 + \cdots + (n-1) = \frac{(n-1)n}{2}.$$

We add  $n$  to both sides to obtain

$$\begin{aligned} 1 + 2 + 3 + \cdots + (n-1) + n &= \frac{(n-1)n}{2} + n \\ &= \frac{n^2 - n + 2n}{2} \\ &= \frac{n^2 + n}{2} \\ &= \frac{n(n+1)}{2}. \end{aligned}$$

Therefore,  $n \in X$ . It follows from the principle of mathematical induction that  $X = \mathbb{N}$ . □

### Exercise 1.2

Prove using mathematical induction that for all positive integers  $n$ ,

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(2n+1)(n+1)}{6}.$$

*Proof.* Let  $X$  be the set of natural numbers from which the formula above holds.  $1 \in X$  because

$$1^2 = \frac{1(2 \cdot 1 + 1)(1 + 1)}{6}.$$

Now suppose that  $n > 1$ , and that  $k \in X$  for all  $k < n$ . In particular,  $n - 1 \in X$  so

$$1^2 + 2^2 + \cdots + (n-1)^2 = \frac{n(n-1)(2n-1)}{6}.$$

Adding  $n^2$  to both sides, we deduce that

$$\begin{aligned}
 1^2 + 2^2 + \cdots + (n-1)^2 + n^2 &= \frac{n(n-1)(2n-1)}{6} + n^2 \\
 &= \frac{2n^3 - 3n^2 + n + 6n^2}{6} \\
 &= \frac{2n^3 + 3n^2 + n}{6} \\
 &= \frac{n(2n^2 + 3n + 1)}{6} \\
 &= \frac{n(2n+1)(n+1)}{6}.
 \end{aligned}$$

So  $n \in X$ , which closes the induction.  $\square$

### Exercise 1.3

Assuming the triangle inequality ( $|x+y| \leq |x|+|y|$ ) is true, show that  $|x_1+x_2+\cdots+x_n| \leq |x_1|+|x_2|+\cdots+|x_n|$  for all  $n \in \mathbb{N}$ .

*Proof.* Let  $X$  be the set of all natural numbers  $n$  such that  $|x_1+x_2+\cdots+x_n| \leq |x_1|+|x_2|+\cdots+|x_n|$ .  $1 \in X$  and it's given that  $2 \in X$ . Suppose that  $k \in X$  for all  $k < n$  where  $n > 2$ . In particular, that means  $n-1 \in X$ . Therefore,  $|x_1+x_2+\cdots+x_{n-1}| \leq |x_1|+|x_2|+\cdots+|x_{n-1}|$ . We add  $|x_n|$  to both sides to obtain

$$|x_1+x_2+\cdots+x_{n-1}|+|x_n| \leq |x_1|+|x_2|+\cdots+|x_{n-1}|+|x_n|.$$

Let  $y = x_1+x_2+\cdots+x_{n-1}$ , then the left-hand side of the inequality above becomes  $|y|+|x_n|$ . Because  $2 \in X$ , we know  $|y+x_n| \leq |y|+|x_n|$ . Therefore,  $|y+x_n| \leq |x_1|+|x_2|+\cdots+|x_n|$ , which closes the induction and shows that  $X = \mathbb{N}$ .  $\square$

### Exercise 1.4

Given a positive integer  $n$ , recall that  $n! = 1 \cdot 2 \cdot 3 \cdots n$  (this is read as  $n$  **factorial**). Provide an inductive definition for  $n!$ . (It is customary to actually use this definition at  $n = 0$ , setting  $0! = 1$ .)

*Proof.*

$$n! = n(n-1)!$$

$\square$

### Exercise 1.5

Prove that  $2^n < n!$  for all  $n \geq 4$ .

*Proof.* Let  $X$  be the set of all natural numbers  $n$  such that  $2^n < n!$ . Define  $S := X \cup \{1, 2, 3\}$ , we show that  $S = \mathbb{N}$  by induction. By definition,  $1 \in S$ . Now suppose that  $k \in S$  for all  $k < n$  where  $n > 1$ . If  $n = 2$  or  $n = 3$ , then by definition  $n \in S$ . If  $n = 4$ , then  $2^4 = 16$  and  $4! = 24$ . Clearly  $16 < 24$ , therefore  $n \in X$ . Now we can solely focus on the cases where  $n > 4$ . We know  $n - 1 \in S$ , so  $2^{n-1} < (n - 1)!$  or  $n - 1 \in \{1, 2, 3\}$ . But  $n > 4$  so  $n - 1 \notin \{1, 2, 3\}$ . So it must be the case that  $2^{n-1} < (n - 1)!$ . Hence,  $2^n < 2(n - 1)!$ . But  $2(n - 1)! < n(n - 1)! = n!$ . Therefore,  $2^n < n!$ . This closes the induction and shows that  $S = \mathbb{N}$ .  $\square$

### Exercise 1.6

Prove that for all positive integers  $n$ ,

$$1^3 + 2^3 + \cdots + n^3 = \left( \frac{n(n+1)}{2} \right)^2.$$

*Proof.* Let  $X$  be the set of all natural numbers  $n$  such that the equality above holds. We show that  $X = \mathbb{N}$  by induction. Because

$$1^3 = \left( \frac{1(1+1)}{2} \right)^2,$$

we know that  $1 \in X$ . Now suppose that  $k \in X$  for all  $k < n$  where  $n > 1$ . In particular,  $n - 1 \in X$ . Therefore,

$$1^3 + 2^3 + \cdots + (n - 1)^3 = \left( \frac{n(n - 1)}{2} \right)^2.$$

Adding  $n^3$  to both sides, we have

$$\begin{aligned}
 1^3 + 2^3 + \cdots + (n-1)^3 + n^3 &= \left( \frac{n(n-1)}{2} \right)^2 + n^3 \\
 &= \frac{n^2(n-1)^2 + 4n^3}{4} \\
 &= \frac{n^2(n^2 - 2n + 1) + 4n^3}{4} \\
 &= \frac{n^4 - 2n^3 + n^2 + 4n^3}{4} \\
 &= \frac{n^4 + 2n^3 + n^2}{4} \\
 &= \frac{n^2(n^2 + 2n + 1)}{2^2} \\
 &= \frac{n^2(n+1)^2}{2^2} \\
 &= \left( \frac{n(n+1)}{2} \right)^2.
 \end{aligned}$$

This closes the induction and proves that  $X = \mathbb{N}$ . □

#### Exercise 1.7

Prove the familiar **geometric progression** formula. Namely, suppose that  $a$  and  $r$  are real numbers with  $r \neq 1$ . Then show that

$$a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a - ar^n}{1 - r}.$$

*Proof.* Let  $S = a + ar + ar^2 + \cdots + ar^{n-1}$ .

$$\begin{aligned}
 S &= a + r(a + ar^2 + \cdots + ar^{n-2}) \\
 &= a + r(S - ar^{n-1}) \\
 &= a + rS - ar^n
 \end{aligned}$$

Thus,  $(1 - r)S = a - ar^n$  and

$$S = \frac{a - ar^n}{1 - r}.$$

□

**Exercise 1.8**

Prove that for all positive integers  $n$ ,

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}.$$

*Proof.* Let  $X$  be the set of natural numbers  $n$  such that the equality above holds. One can quickly verify that  $1 \in X$ . Now suppose that  $k \in X$  for all  $k < n$  where  $n > 1$ . In particular,  $n-1 \in X$ . Hence,

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n-1)} = \frac{n-1}{n}.$$

Adding  $\frac{1}{n(n+1)}$  to both sides and performing some simple algebra, we yield

$$\begin{aligned} \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} &= \frac{n-1}{n} + \frac{1}{n(n+1)} \\ &= \frac{(n-1)(n+1) + 1}{n(n+1)} \\ &= \frac{n^2 - 1 + 1}{n(n+1)} \\ &= \frac{n^2}{n(n+1)} &= \frac{n}{n+1}. \end{aligned}$$

This closes the induction and shows that  $X = \mathbb{N}$ . □

**Exercise 1.9**

By trial and error, try to find the smallest positive integer expressible as  $12x + 28y$ , where  $x$  and  $y$  are allowed to be any integers.

*Proof.*  $x = -2$  and  $y = 1$  gives 4. □

**Exercise 1.10**

Use mathematical induction to show that the complete graph of  $n$  points has exactly  $n(n-1)/2$  lines.

*Proof.* Let  $X$  be the set of natural numbers  $n$  for which the formula above works. We show that  $S := X \cup \{1\}$  is equal to  $\mathbb{N}$  by induction. By definition,  $1 \in S$ . Suppose that  $k \in S$  for all  $k < n$  where  $n > 1$ . Assume  $n = 2$ , then  $2(2-1)/2 = 1$ , so  $2 \in S$ . Assume  $n > 2$ , we know from our induction hypothesis that  $n-1 \in S$ , but  $n-1 > 1$  and thus cannot be an element of  $\{1\}$ . So  $n-1 \in X$ . So the complete graph of  $n-1$  has exactly  $(n-1)(n-2)/2$  lines. Now add another point to the graph. That point will form a line with every other point (of which there are  $n-1$ ). So

$n - 1$  new lines will be formed. Hence, the complete graph of  $n$  will have  $(n - 1)(n - 2)/2 + (n - 1)$  lines. Using some algebra,

$$\begin{aligned}\frac{(n - 1)(n - 2) + 2(n - 1)}{2} &= \frac{(n - 1)(n - 2 + 2)}{2} \\ &= \frac{n(n - 1)}{2}.\end{aligned}$$

This closes the induction and shows that  $S = \mathbb{N}$ , as desired. □